Review

Mathematical Statistics

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Conceptual review

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- Sample mean and variance

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- Sample mean and variance
- Conditional mean

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- Sample mean and variance
- Conditional mean
- Regression

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- Sample mean and variance
- Conditional mean
- Regression
- Average treatment effect

Conceptual review

- Covariance and correlation coefficient

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- Correlation vs causation

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- Correlation vs causation
- Bias and standard error

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- Correlation vs causation
- Bias and standard error
- Confidence intervals

Conceptual review

- Covariance and correlation coefficient
- Correlation vs causation
- Bias and standard error
- Confidence intervals
- The bootstrap

Weight-loss supplement

In order to evaluate the effectiveness of a weight-loss supplement, we record the weight gain/loss in kilograms after one month of a treatment group that takes the supplement, and a control group that doesn't.

In bold, we highlight the participants who report that they exercise daily.

Observed average treatment effect?

Supplement: 0.5, **-4.6**, **-7.5**, 1.1, **-3.4**, **-1.4**, **3.2**, -1.9, **-3.6**, 2.4, 1.2, **4.2**, **-2.5**, **-3.5**, 4.7

No supplement: 1.6, **-6.7**, 2.3, **-4.5**, **3.4**, -0.6, 2.3, 1.5, 1.7, 2.6, 0.5, 1.2, **-3.6**, 2.1, -1.2, 0.4

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unadjusted ATE

$$= \frac{0.5 - 4.6 - 7.5 + 1.1 - 3.4 - 1.4 + 3.2 - 1.9 - 3.6 + 2.4 + 1.2 + 4.2 - 2.5 - 3.5 + 4.7}{15}$$

$$- \frac{1.6 - 6.7 + 2.3 - 4.5 + 3.4 - 0.6 + 2.3 + 1.5 + 1.7 + 2.6 + 0.5 + 1.2 - 3.6 + 2.1 - 1.2 + 0.4}{16}$$

$$= -0.93$$

Average treatment effect?

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$$
\begin{aligned}
p_{\tilde{x}}(1) &= \frac{13}{31} = 0.42, \\
\mu_{\tilde{w} \mid \tilde{t},\tilde{x}}(0,0) &= \frac{1.6 + 2.3 - 0.6 + 2.3 + 1.5 + 1.7 + 2.6 + 0.5 + 1.2 + 2.1 - 1.2 + 0.4}{12} \\
&= 1.2, \\
\mu_{\tilde{w} \mid \tilde{t},\tilde{x}}(0,1) &= \frac{-6.7 - 4.5 + 3.4 - 3.6}{4} = -2.85, \\
\mu_{\tilde{w} \mid \tilde{t},\tilde{x}}(1,0) &= \frac{0.5 + 1.1 - 1.9 + 2.4 + 1.2 + 4.7}{6} = 1.33, \\
\mu_{\tilde{w} \mid \tilde{t},\tilde{x}}(1,1) &= \frac{-4.6 - 7.5 - 3.4 - 1.4 + 3.2 - 3.6 + 4.2 - 2.5 - 3.5}{9} = -2.12,
\end{aligned}
$$

Average treatment effect?

Average treatment effect?

$$
\begin{aligned}
\mathrm{E}\left[\widetilde{\mathrm{po}}_1\right] &= \sum_{x=0}^{1} p_{\tilde{x}}(x)\mu_{\widetilde{\mathrm{po}}_1 \mid \tilde{x}}(x) \\
&= \sum_{x=0}^{1} p_{\tilde{x}}(x)\mu_{\widetilde{\mathrm{po}}_1 \mid \tilde{x},\tilde{t}}(1,x) \\
&= \sum_{x=0}^{1} p_{\tilde{x}}(x)\mu_{\tilde{w} \mid \tilde{x},\tilde{t}}(1,x) \\
&= 0.58 \cdot 1.33 - 0.42 \cdot 2.12 = -0.119, \\
\mathrm{E}\left[\widetilde{\mathrm{po}}_0\right] &= \sum_{x=0}^{1} p_{\tilde{x}}(x)\mu_{\tilde{w} \mid \tilde{x},\tilde{t}}(0,x) \\
&= 0.58 \cdot 1.2 - 0.42 \cdot 2.85 = -0.501 \\
\mathrm{ATE} &= \mathrm{E}\left[\widetilde{\mathrm{po}}_1\right] - \mathrm{E}\left[\widetilde{\mathrm{po}}_0\right] = 0.382
\end{aligned}
$$

Three random variables

If a random variable $\tilde{w}$ is positively correlated with another random variable $\tilde{y}$, and $\tilde{y}$ is positively correlated with a third random variable $\tilde{z}$, can $\tilde{w}$ and $\tilde{z}$ be negatively correlated? Provide an example of three such variables, or prove that it is impossible.

Three random variables

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Hint: For random variables $\tilde{a}$, $\tilde{b}$ and $\tilde{c}$ with zero mean and unit variance, which are pairwise uncorrelated:

$$\begin{aligned}\tilde{w} &:= \tilde{a} + \alpha\tilde{b},\\ \tilde{y} &:= \tilde{b} + \beta\tilde{c},\\ \tilde{z} &:= \gamma\tilde{a} + \tilde{c},\end{aligned}$$

Three random variables

$$\tilde{w} := \tilde{a} + \alpha\tilde{b}, \quad \tilde{y} := \tilde{b} + \beta\tilde{c}, \quad \tilde{z} := \gamma\tilde{a} + \tilde{c},$$

Three random variables

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$$\begin{aligned} \mathrm{E}[\tilde{w}] &= \mathrm{E}[\tilde{a}] + \alpha\mathrm{E}[\tilde{b}] = 0, \\ \mathrm{E}[\tilde{y}] &= \mathrm{E}[\tilde{b}] + \beta\mathrm{E}[\tilde{c}] = 0, \\ \mathrm{E}[\tilde{z}] &= \gamma\mathrm{E}[\tilde{a}] + \mathrm{E}[\tilde{c}] = 0. \end{aligned}$$

Since $\tilde{a}$, $\tilde{b}$ and $\tilde{c}$ have zero mean and are pairwise uncorrelated,

$$\mathrm{E}[\tilde{a}\tilde{b}] = \mathrm{E}[\tilde{a}\tilde{c}] = \mathrm{E}[\tilde{b}\tilde{c}] = 0.$$

Three random variables

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Three random variables

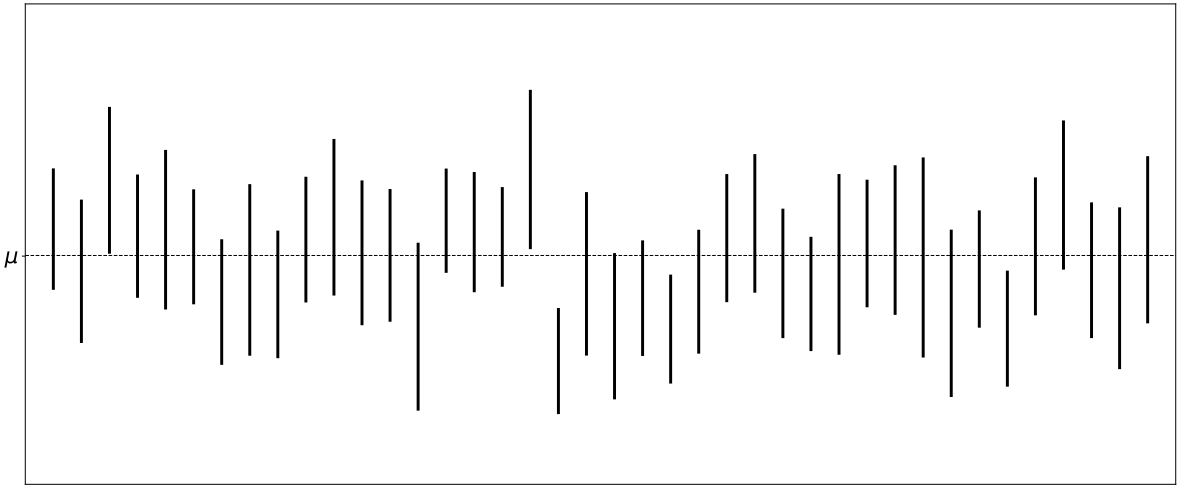
$$\tilde{w} := \tilde{a} + \alpha\tilde{b}, \quad \tilde{y} := \tilde{b} + \beta\tilde{c}, \quad \tilde{z} := \gamma\tilde{a} + \tilde{c},$$

By linearity of expectation,

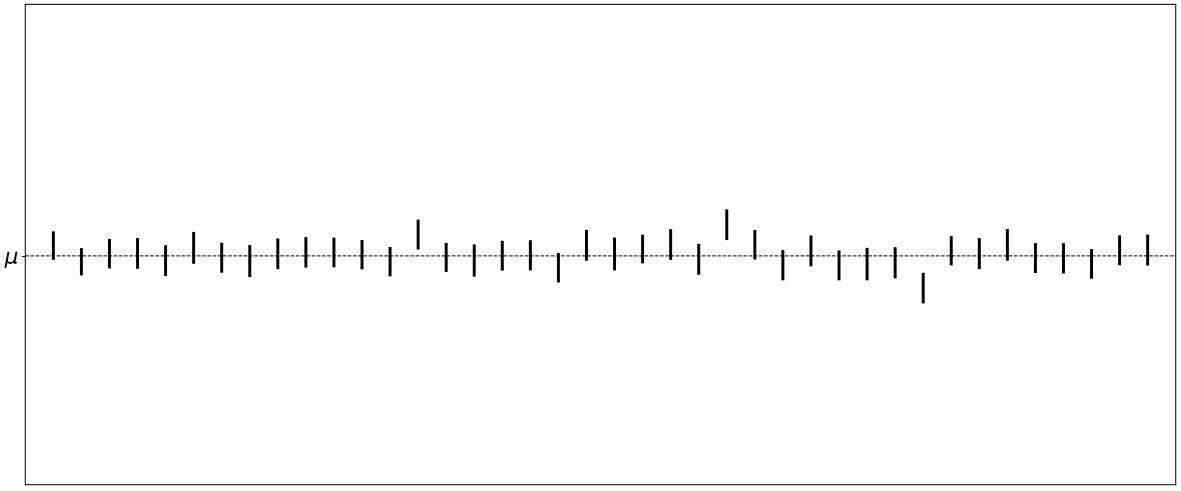
$$\begin{aligned}
\mathrm{E}\left[\tilde{w}\tilde{y}\right] &= \mathrm{E}[\tilde{a}\tilde{b}] + \beta\mathrm{E}\left[\tilde{a}\tilde{c}\right] + \alpha\mathrm{E}[\tilde{b}^2] + \alpha\beta\mathrm{E}[\tilde{b}\tilde{c}] \\
&= \alpha, \\
\mathrm{E}\left[\tilde{w}\tilde{z}\right] &= \gamma\mathrm{E}\left[\tilde{a}^2\right] + \mathrm{E}\left[\tilde{a}\tilde{c}\right] + \gamma\alpha\mathrm{E}[\tilde{a}\tilde{b}] + \alpha\mathrm{E}[\tilde{b}\tilde{c}] \\
&= \gamma, \\
\mathrm{E}\left[\tilde{y}\tilde{z}\right] &= \gamma\mathrm{E}[\tilde{a}\tilde{b}] + \mathrm{E}[\tilde{b}\tilde{c}] + \gamma\beta\mathrm{E}\left[\tilde{a}\tilde{c}\right] + \beta\mathrm{E}\left[\tilde{c}^2\right] \\
&= \beta.
\end{aligned}$$

$1 - \alpha$ confidence intervals for μ

A

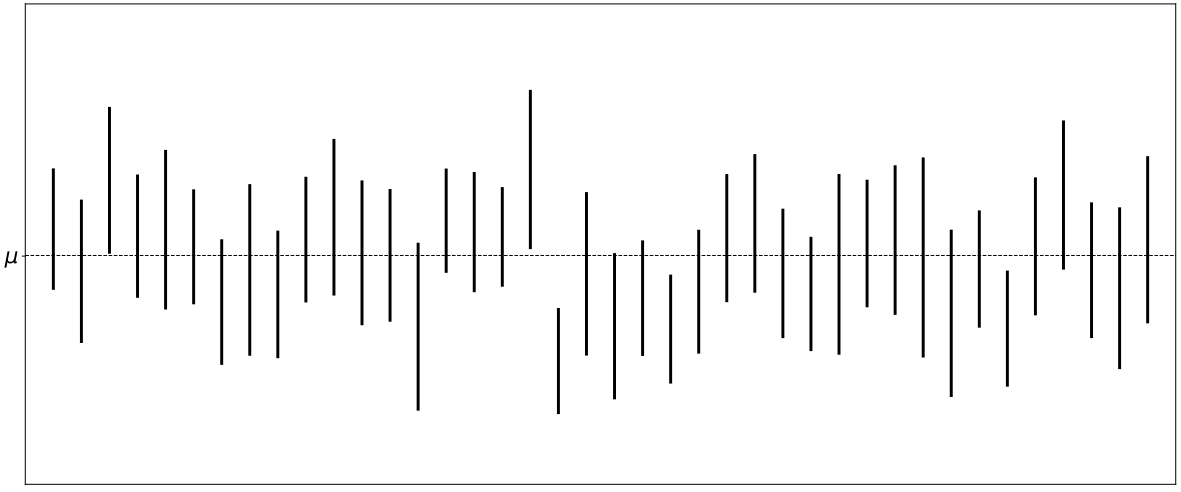


B

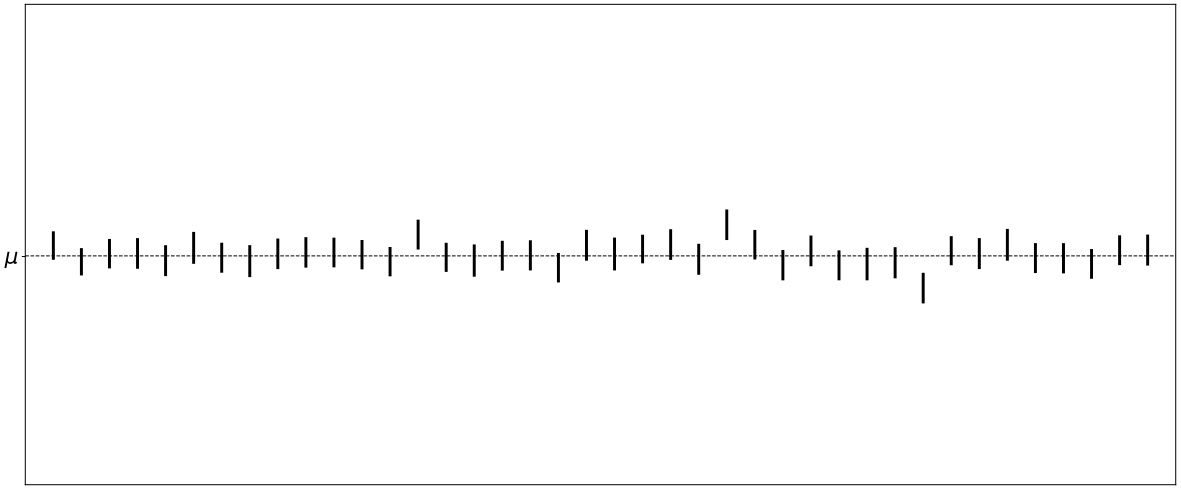


CLT or Chebyshev's inequality?

A

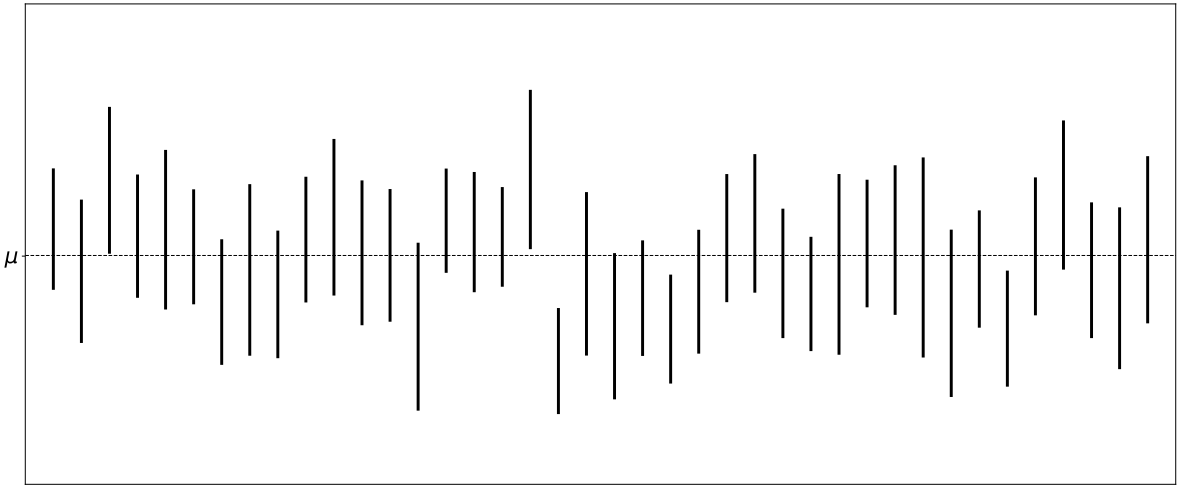


B

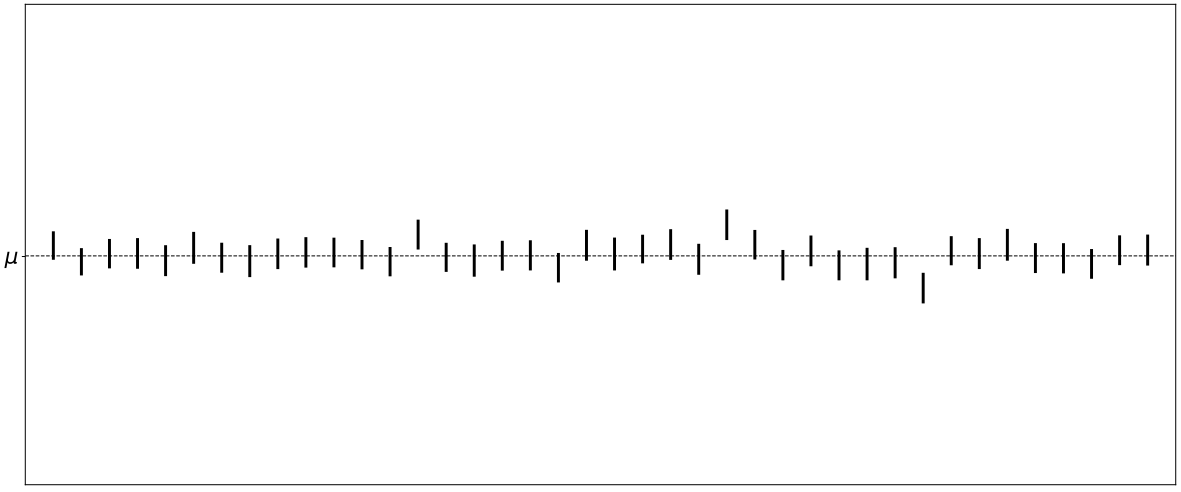


More data?

A

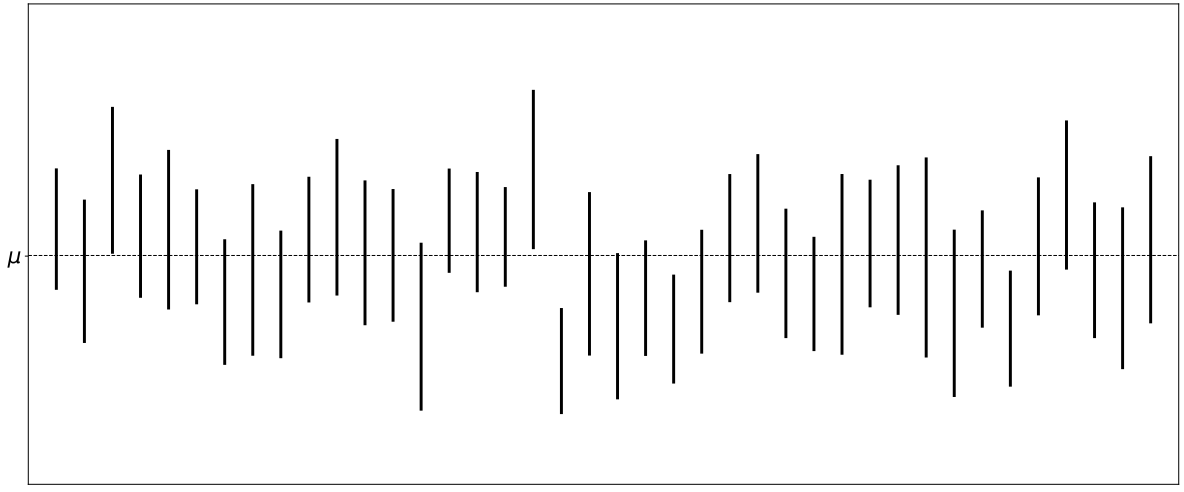


B

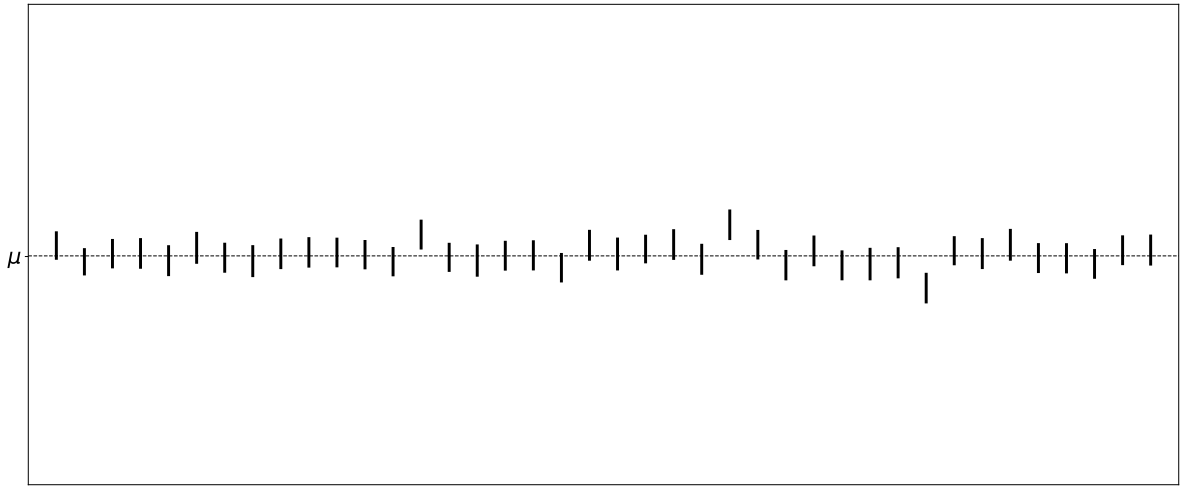


Higher population variance?

A



B



Cholesterol

200 men and 100 women sampled independently at random with replacement from the population. We assume the population standard deviations of cholesterol are known: σ_{men}, σ_{women}. Standard error of the difference between the average cholesterol in men and women?

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By independence

$$\begin{aligned}
\operatorname{Var}\left[\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i - \frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i\right] &= \operatorname{Var}\left[\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i\right] + \operatorname{Var}\left[-\frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i\right] \\
&= \operatorname{Var}\left[\frac{1}{200}\sum_{i=1}^{200}\tilde{x}_i\right] + \operatorname{Var}\left[\frac{1}{100}\sum_{i=1}^{100}\tilde{y}_i\right] \\
&= \frac{\sigma_{\text{men}}^2}{200} + \frac{\sigma_{\text{women}}^2}{100}
\end{aligned}$$

Cholesterol

Average cholesterol: 200 for the men, 180 for the women

$\sigma_{\text{men}} = \sigma_{\text{women}} = 10$

95% confidence interval for the difference between the mean cholesterol level of men and women?

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95% confidence interval for the difference between the mean cholesterol level of men and women?

By CLT, each average is Gaussian, so difference is Gaussian with mean $\mu_{\text{men}} - \mu_{\text{women}}$ and variance $\frac{\sigma^2_{\text{men}}}{200} + \frac{\sigma^2_{\text{women}}}{100} = 1.5$. Consequently, a 0.95 confidence interval for $\mu_{\text{men}} - \mu_{\text{women}}$ is

$$\left[20 - 1.96\sqrt{1.5}, 20 + 1.96\sqrt{1.5}\right] = [17.6, 22.4].$$

Modified Gaussian bootstrap confidence interval

Bootstrap percentile confidence intervals work for estimators that are rendered approximately Gaussian by a strictly monotonic transformation, even if the transformation is not known to us. Propose an alternative approach, based on the bootstrap standard error, to build confidence intervals when the transformation is known (as in the case of Fisher's transformation)